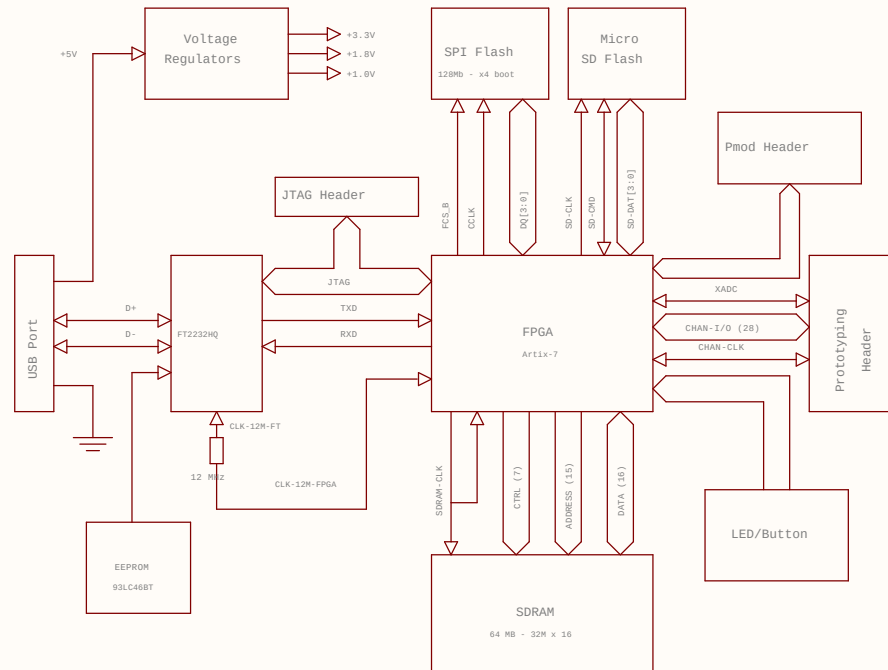
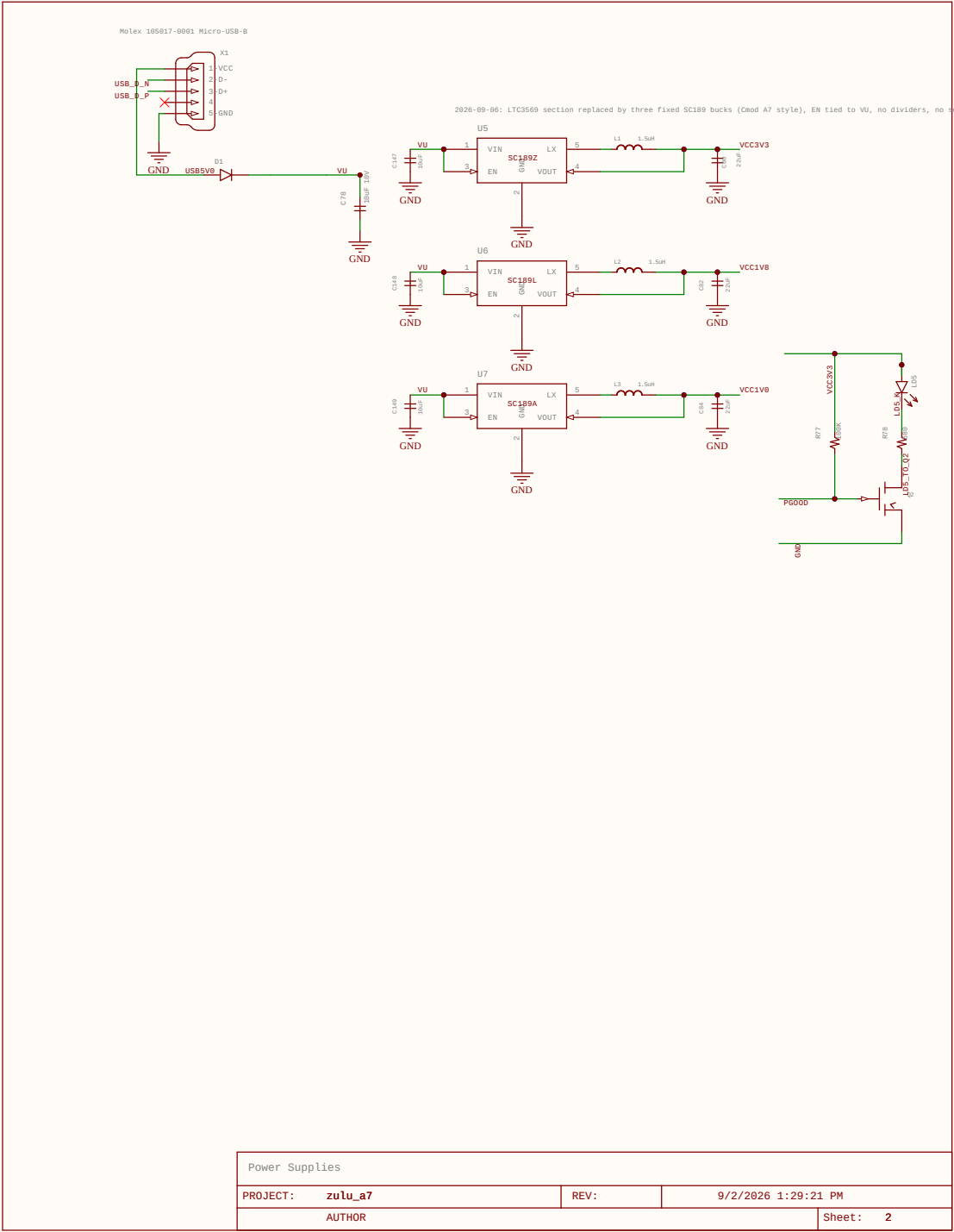


SPI flash sits on the FPGA's dedicated master-SPI config pins: CCLK C11, FCS_B K19, DQ[3:0] on D18/D19/G18/F18. Read at power-up.

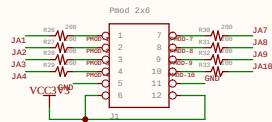
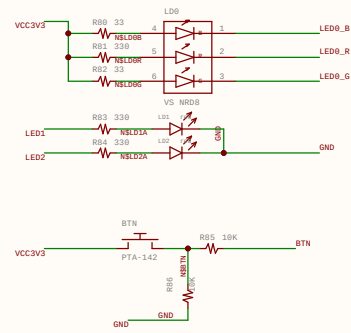
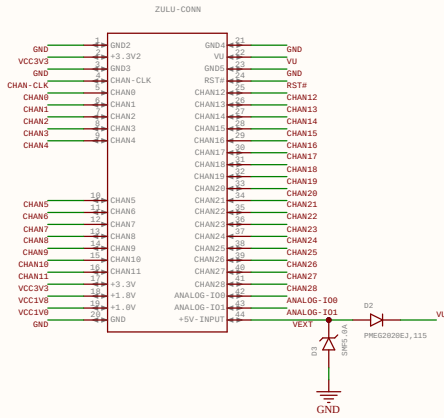
microSD runs the native 4-bit SD bus, not SPI: CLK, CMD and DAT[3:0] on six bank-34 balls of its own, so a card cannot touch the config bus.



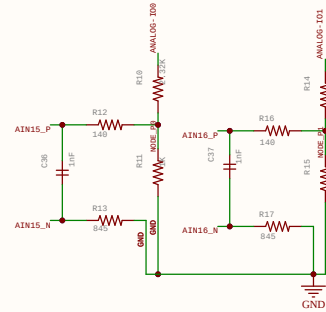
Block Diagram			
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X2: 44 pins on a 2.54 mm grid, rows 0.700 in apart, board 69.85 x 20.32 mm.
Numbered right to left and straight through: top row 1-20, bottom row 21-44.
TOP ROW, x 50.69 down to 1.27: GND, +3.3V, GND, then CHAN-CLK and CHAN0..CHAN3
(pins 4-8), CHAN4 (pin 9), CHAN5..CHAN11 (10-16), +3.3V, +1.8V, +1.0V, GND.
FOUR positions between pin 9 and pin 10 -- x 26.82, 34.29, 31.75, 29.21 -- are
RESERVED, not spare: the landing for the edge-mounted micro-USB X1, 7.80 mm of
copper in 11.176 mm clear, centred with 1.69 mm each side.
BOTTOM ROW, contiguous, x 50.69 down to 1.27: GND, VU, GND, RST#, CHAN12..CHAN28
(pins 25-41), ANALOG-I00, ANALOG-I01, +5V-INPUT.
FIVE grounds: pins 1, 9, 20, 21, 23. Pin 22-VU SOURCES 5 V, pin 44 only takes it.



Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are W16/W27/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



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General IO

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U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16SA-6TIN it replaces -- all 54 pins and every
TSP: 1 dimension checked against both datasheets, and
pin 40 is NC on both.
One DIFFERENCE, and IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2026-09-08 (-6TIN dry): 143 MHz max at CL13,
tRCD/trp 21, tRC 63 ns, TA 0-70 C, IOD1 max 110 mA.

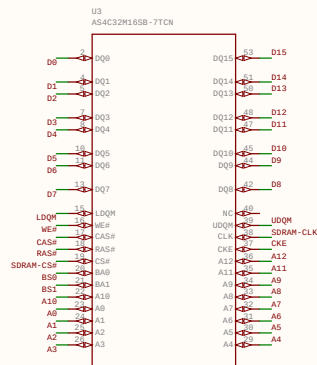
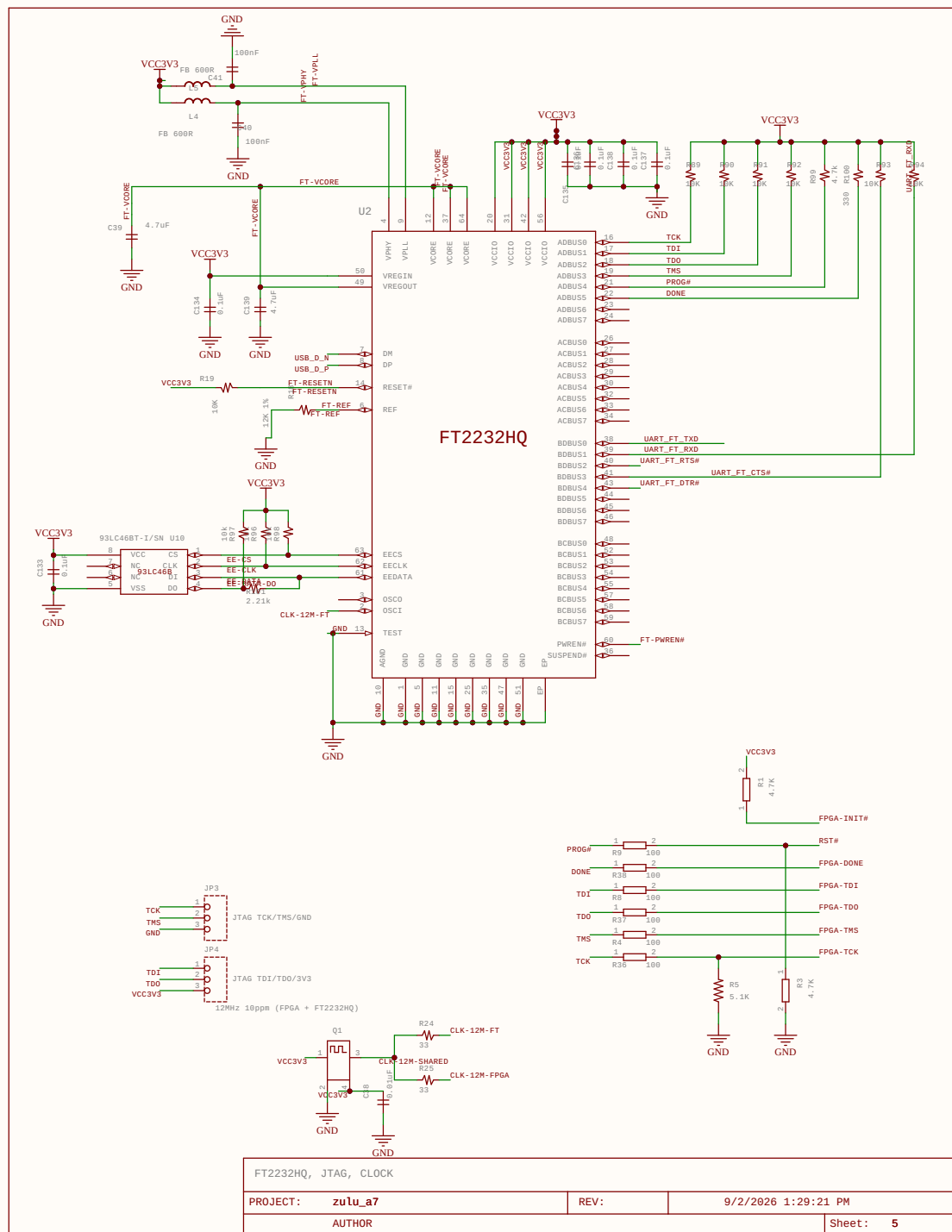
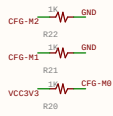


Figure 1 shows a schematic diagram of an 8-bit DAC. It consists of eight stages, each with a 1-bit input (VSSQ[3:0] to VSSQ[9:8]) and a 1-bit output (VDDQ[3:0] to VDDQ[9:8]). Each stage contains a 1-bit DAC core (VCC3V3, 1.2uF, 1.2uF) and a 1-bit DAC core (VCC3V3, 1.2uF, 1.2uF). The outputs are summed to produce the final 8-bit DAC output.

Memory			
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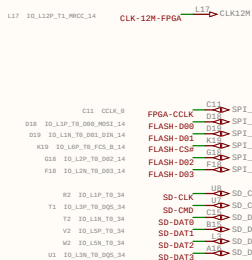


PROTOTYPING HEADER



CLK

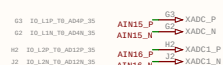
FLASH



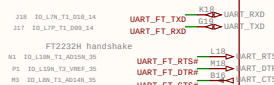
JTAG



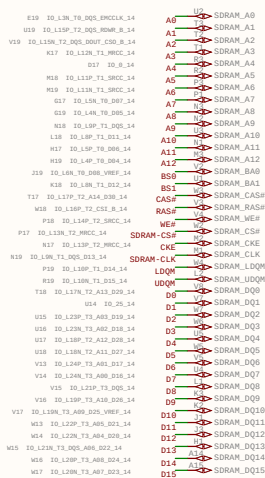
ANALOG



FT2232H



SDRAM



UNCONNECTED BALLS
XADC analog (unused)



Thermal diode (unused)



Battery backup (unused)



GTP - float (UG482 TS-4)



D1 MGTPTXND_216



ES MGTREFCLKSP_216



B10 MGTFECLKIP_216



A2 MGTPXN1_216



Pmod JA (bank 16 x5)
A14 IO L6P TR 16



```

A15  IO_LEN_TO_VREF_16
B16  TO_1316_V1_SWCC_16

```



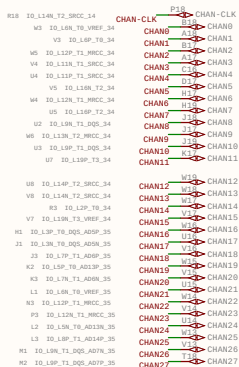
A16 IO_L12P_Y1_M0CC_16



W19 IO_L16N_T2_A15_031_14

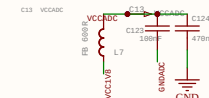
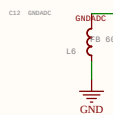


W7 IO_L1SP_T2_MRCC_34

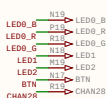


The six pins above CCLK₁ are the ONLY halves master-SPI configuration can use (UCG470). They are the ordinary user I/O once DONE goes high. SDRAM A4 master-SPI → D17 to free FCS_B; the ESP reset moved D17 → B15. microSD is on its own CLK/CMD/DAT0 so a card cannot fight the configuration bus.

The six microSD signals below CCLK/Flash are the native 4-bit SD bus: CLK CMD DAT0-DAT3. All sit in bank 34 byte group 10 (S2 I17 to U3 V2 W0). DAT3/DAT2 look CHAN1/CHAN0 off the prototyping header; DAT3 is the SD SPI chip-select, rendered, only two pins were spent.



LED / BUTTON



FPGA Connections			
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